LEARNER GUIDE

Combining the theorems

Circle Theorems · Lesson 9. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

By now you own a full toolkit of circle theorems. Real problems rarely use just one — they need **several, in sequence**. The technique is the **angle chase**: find any angle you can reach from the given facts, **write the reason**, and use that new angle to unlock the next, repeating until you have what you need. The single most useful habit is to **name the theorem at every step** — that is what turns a string of numbers into a watertight argument. A good first move is to look for the easiest angle you can already reach: a central angle to halve, a diameter giving a right angle, or two angles standing on the same chord.

What you are learning to notice

- A multi-step problem is solved **one angle at a time**.
- Each answer becomes a **tool** for the next step.
- **Name the theorem** at every step — the chain of reasons is the solution.
- Choose a productive **first move**: something you can reach from the givens.

Moving through it, one step at a time

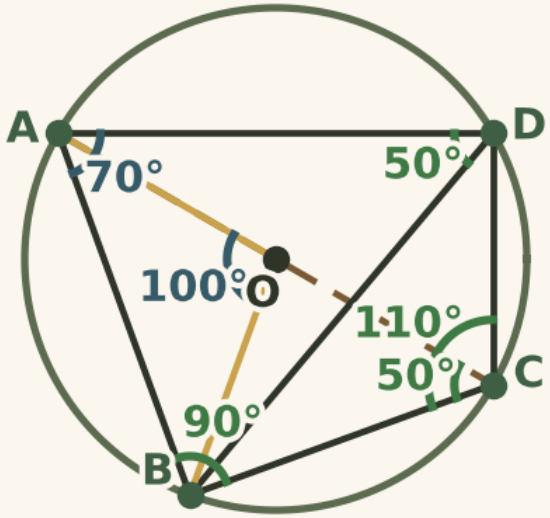
Read the givens and mark everything on the diagram. Find one angle you can reach and label it with its reason. Look for what that new angle unlocks — often an isosceles triangle (equal radii), a diameter (right angle), a same-segment partner, or a cyclic-quadrilateral opposite angle. Keep going, one reasoned step at a time, until you reach the target.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — A three-step chase

In the circle, O is the centre, AC is a diameter, and the central angle ∠AOB = 100°. Find ∠ACB, ∠ABC and ∠ADB, giving a reason for each.

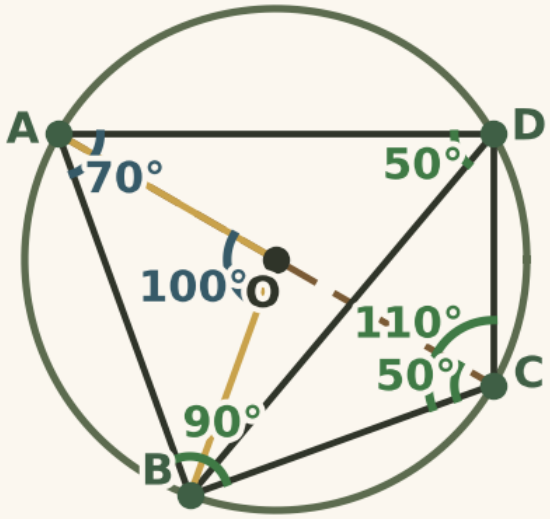


1. ∠ACB and ∠AOB stand on the same arc AB; the angle at the centre is twice the angle at the circumference.	∠ACB = 100 ÷ 2 = 50°
2. AC passes through O, so AC is a diameter; the angle in a semicircle is 90°.	∠ABC = 90°
3. ∠ADB and ∠ACB stand on the same chord AB from the same segment.	∠ADB = ∠ACB = 50°

∠ACB = 50°, ∠ABC = 90°, ∠ADB = 50° — each justified by a named theorem.

Example 2 — Extending the chase with a cyclic quadrilateral

In the same figure, ABCD is a cyclic quadrilateral and ∠BAD = 70°. Find the opposite angle ∠BCD.



1. ABCD is a cyclic quadrilateral, so its opposite angles add to 180°.	∠BCD + ∠BAD = 180°
2. Substitute the known angle ∠BAD = 70°.	∠BCD = 180 – 70

∠BCD = 110° — opposite angles of a cyclic quadrilateral are supplementary.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (all the circle theorems so far: angle at the centre, angle in a semicircle, angles in the same segment, cyclic quadrilaterals, tangents, and the alternate segment theorem), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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